Wenxuan Wang

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Homepages: https://amazing-wx.github.io

Research interest: Design and motion control of parallel robot

EDUCATION

Harbin Institute of Technology, Shenzhen (HITSZ) **Shen Zhen, China**

Master of Mechanical Engineering **Sept. 2023 – Mar.2026**

- Supervisor: Prof. Bing Li (National “Ten Thousand People Project”)
- Relevant Courses: Numerical Analysis, Modern Control Theory, Collaborative Robotics Technology and Applications, etc.

Lanzhou University of Technology **Lan Zhou, China**

Bachelor of Mechanical Design, Manufacturing, and Automation **Sept. 2019 – Jun.2023**

- **GPA: 91.55**|100 Ranking **0.6%** (1/459) **National Scholarship for twice**、National Encouragement Scholarship
- Relevant Courses: Linear Algebra (100), Advanced Mathematics (95), Mechanics of Material (99), Mechanical Design (97) Electronic technology (96), Mechanical Principles (95), Calculation Method (96), Basic Mechanic Engineering Control (94)

HONORS AND AWARDS

- 100 **Outstanding National Scholarship Representatives** (only 100 a year in China) |Published in People's Daily **May. 2022**
- **National Scholarship (Top 1, ¥8000, 0.2%)** **2019-2020**
- **National Scholarship (Top 1, ¥8000, 0.2%)** **2020-2021**
- **National Encouragement Scholarship (Top 10%, ¥5000)** **2021-2022**
- The First Prize Scholarship **for three times** **2020, 2021, 2022**
- **Provincial-Level Merit Student** (**only 2** in the same grade) **2022**
- Merit Student / Pacemaker to Merit Student **2020, 2021/2022**
- **Outstanding Graduates** in Lanzhou University of Technology **2023**
- The Second Graduate Scholarship of HITSZ **2023, 2024**

PUBLICATIONS AND ON-GONING

- **Wenxuan Wang,** Yinghao Ning, Yang Zhang, Peng Xu*, Bing Li*. Linear active disturbance rejection control with linear quadratic regulator for Stewart platform in active wave compensation system. ***Applied Ocean Research*.2025**. (**JCR Q1. IF=4.3)**
- Yang Zhang, **Wenxuan Wang**, Xi Kang*, Bing Li*. Design, analysis, and experimentation of deployable multi-closed-loop truss modules for aerospace platforms. ***Mechanism and Machine Theory*, 2025 (JCR Q1. IF=4.5)**
- **Wenxuan Wang**, Xiaokai Cui, Peng Xu*, Bing Li*. An improved linear quadratic regulator of shipborne Stewart platform for wave compensation, ***IEEE International Conference on Robotics and Biomimetics***. **Bangkok, Thailand, 2024 (EI).**

RESEARCH EXPERIENCE

Research on Design and Motion Control of a Six-DOF Parallel Stabilized Platform for Offshore Wave Compensation | Key number of Commissioned by China General Nuclear Power Group Laboratory| **Dec.2023--Present**

- Designed a doubled-motion platform based on the 6UCU Stewart parallel mechanism, where the bottom platform simulates wave motion and the top platform is stabilized by controlling the length of six limbs to isolate the vibration of waves. Completed the forward, inverse kinematic solution, workspace analysis, and control of 6UCU Stewart parallel.
- Proposed an improved linear active disturbance reject control (ILADRC) based on linear quadratic regulator (LQR) in joint space for each chain electric-driven actuator (PMSM) to achieve accurate stabilization of the top platform.
- Utilized LQR to replace the series PD control of the traditional LADRC to solve the problem of controller bandwidth. Used the LESO to estimate and compensate for the total disturbance of system.
- Verified the proposed controller ILADRC with the Lyapunov theory. Solved the problem of overshooting and oscillation of traditional PI control, and improved the anti-disturbance ability of the system.
- Project Video: https://www.youtube.com/watch?v=jiMR-fIs53g https://youtu.be/rz-4keuo-h4

Research on hybrid variable stiffness collaborative robot for abrasive machining with ILADRC algorithm| Key member of 6th China Graduate Student Robot Innovation and Design Competition **Nov. 2023 – Dec. 2024**

- Proposed a novel 3-DOF redundant parallel end-effector mechanism (2UPR-2RRU) with two rotational and one translational degrees of freedom, incorporating a lightweight structural design for enhanced performance.

- Developed an improved Linear Active Disturbance Rejection Control (ILADRC) algorithm that uses an exponential function for adaptive parameter adjustment to compensate for position errors caused by unknown disturbances, thereby enabling online tuning of ADRC parameters based on tracking errors and resulting in significantly improved control performance.
- Designed a fuzzy-adaptive impedance control system, incorporating a fuzzy logic loop into the adaptive term to formulate a fuzzy conductance controller, effectively mitigating force overshoot during grinding force tracking.

Achievement: National Third Prize of the 6th China Graduate Student Robot Innovation and Design Competition

- Project Video: https://youtu.be/61c5QmwmCeU?si=X7ylEsiabiaAcuwA

National Undergraduate Training Program for Innovation and Entrepreneurship: Mechanical Design and Control of a New Traffic Directing Robot |(Project Number:202210731019)----Project Leader **Sept. 2022—Jun. 2023**

- Designed an intelligent traffic-directing robot based on an STM32 microcontroller, equipped with dual 7-DOF humanoid arms and four Mecanum wheels for omnidirectional mobility, enabling autonomous traffic control and mobile law enforcement.
- Programmed precise servo control through PWM duty cycle modulation, allowing the robotic arms to accurately replicate standard traffic police hand signals.
- Integrated onboard cameras and image recognition algorithms with the mobile platform to identify traffic violations, addressing the limitations of fixed-location surveillance systems.
- **Achievement: National First Prize** of the 23rd China Robotics and Artificial Intelligence Competition.
  National First Prize of International Youth Artificial Intelligence Innovation Competition.
- Project Video: https://youtu.be/FKRFKs0Cmow

National Undergraduate Science and Technology Innovation Project: Motion Control of a Smart Home Security Robot | (Project Number:202110731012) ----Project Leader **Oct. 2021—Jun. 2022**

- Responsible for the mechanical design, simulation construction, and motion control of robotics.
- Completed the control of the robot chassis movement and the grasping action of the 5-DOF arm with Arduino chip.
- Utilized ultrasonic sensors and Arduino to communicate with each other through Usart. Adjusted the baud rate communication to achieve the obstacle avoidance function.
- **Achievement**: **National Second Prize** of the 3rd China University Intelligent Robot Creative Competition.
- Project Video: https://youtu.be/VATWh5RnMGQ

CAMPUS EXPERIENCE

Secretary of the Youth League Branch Committee | Undergraduate **Sept. 2019- Jun. 2023**

- Organized and planned **30+** class league activities, achieving a participation rate of over **95%**, effectively enhancing class cohesion. Led the class in school-wide and community volunteer activities, with 50+ student participation.
- Spearheaded and participated in the planning and execution winning **2 first-place awards** in university-level competitions. The class league branch was awarded the prestigious "**Top 100** Provincial Outstanding League Branch" for its vitality and excellence

Vice-President of the '3J' Science, Technology and Innovation Association **Sept. 2020- Sept. 2021**

- Led the planning and execution of **5+** major campus-wide technology innovation competitions, providing technical support and pre-competition, and attracting a total of over **400** students to participate in competitions.
- Coached and trained students for various competitions, resulting in numerous members achieving top placements in university-level contests. Successfully contributed to the association receiving the prestigious "**Top 10 University Association**" award.

COMPETITIONS AND AWARDS

- 1 **National First Prize** of the 23rd China Robotics and Artificial Intelligence Competition. **Dec.2021**
- 2 **National First Prize** of the International Youth Artificial Intelligence Innovation Competition. **Nov.2021**
- 3 **National First Prize** of the 14th National 3D Digital Innovation Design Competition. **Aug.2021**
- 4 **National Second Prize** of the 12th Process Equipment Practice and Innovation Competition. **Dec.2021**
- 5 **National Second Prize** of the 17th Embedded Artificial Intelligence Competition for Students **Dec.2021**
- 6 **National Second Prize** of the 3rd China University Intelligent Robot Creative Competition. **Dec.2020**
- 7 **Bronze Award** of the 7th China College Students "Internet+" Innovation and Entrepreneurship Competition **Dec.2021**
- 8 **First Prize** at the provincial level of the Chinese Mathematics Competitions **Dec.2020**
- 9 **First Prize** at the provincial level of the China Undergraduate Mathematical Contest in Modeling **Nov.2020**
- 10 **Honorable Mention** of Mathematical Contest in Modeling **Apr.2021**

SKILLS

- **Modeling:** Experience with Matlab, C/C++, Python; Ansys, Adams, Solidworks, AutoCAD, Photoshop, etc.
- **Language:** English (Fluent, CET-4/6); IELTS.